

Autonomous robotics and reality capture partnership

PSB Academy × OpenSpace

1. Executive summary

Singapore's transformation of its built environment is, at its core, a productivity goal: to build more while depending on fewer people on site. The Building and Construction Authority (BCA) pursues this through its Integrated Digital Delivery (IDD) framework and, within it, ops-tech-enabled inspections — where 360-degree reality capture replaces physical site visits. The results are already material: BCA reports productivity savings of up to 60% in the inspection process on projects that adopt virtual inspection, as officers no longer need to travel to and walk each site in person.

Captured consistently from groundbreaking through to handover, that same imagery also becomes a continuous record of the project — supporting progress tracking, quality control, and safety during construction, and leaving the project ready for a virtual Temporary Occupation Permit (TOP) inspection at completion. One manual step still remains, however. Today the capture itself is a person walking the site with a 360 camera. That is the last labour-dependent link in the chain, and the one this partnership is built to remove.

Building on successful virtual TOP pilots with BCA, BCA now mandates reality capture and virtual inspection across its construction projects. BCA, in turn, has embedded 360-degree capture into its TOP/CSC inspection framework and is rolling it out in phases by building type. For the wider industry, the direction of travel is clear.

This proposal positions a partnership between PSB Academy and OpenSpace within that national direction. By pairing the OpenSpace 360-degree capture platform and its API ecosystem with autonomous robotics, we aim to take people out of the routine capture loop entirely: a robot performs the daily site walk, captures the data, and uploads it for remote review without manual effort — freeing skilled staff from repetitive documentation for higher-value work. Beyond the technology, the partnership is built to develop the engineers who will lead reality-capture and robotics adoption across the Asia Pacific region.

From remote inspection to autonomous capture. Virtual inspection already removed the need for officers to be on site. The next productivity step is removing the manual walk

itself — letting a robot capture the site consistently and repeatedly, so skilled people are freed from routine documentation rather than tied to it.

2. Problem statement: the reality capture challenge

The industry still treats site recording as a tedious manual walk, a task often handed to graduates and cadets as their introduction to the field. The process is inefficient, prone to human error, and vulnerable to shortcuts where areas get missed — which undermines the integrity of the project record and, ultimately, the inspection that depends on it.

Robotic hardware and reality-capture software are both commercially available, but integrating them for live, dynamic construction sites remains an unsolved problem. The full integration challenge breaks down into a set of roughly twenty discrete problems that must be solved in unison; the most critical are:

- **Integrated autonomy:** navigation logic that avoids obstacles and prevents the shortcut routes a human walker might take.
- **Safety responsiveness:** hardware that can detect and respond to site cues such as sirens or emergency alarms.
- **Power management:** autonomous return-to-base for independent battery charging.
- **Hardware integration:** mounting a high-precision 360-degree camera with appropriate vibration dampening.

3. Project objectives

The partnership pursues a dual-purpose mission that balances rigorous industry requirements with our institutional focus on student development:

- **Educational empowerment:** give students a platform for entrepreneurship. We are candid that the student journey comes first — the priority is to give students a competitive edge by benchmarking their work against world-class AI and robotics standards.
- **Technical innovation:** build a fully integrated autonomous solution in which a robot can complete daily site walks, capture data, and upload it through the OpenSpace APIs without human intervention.
- **Industry and regulatory validation:** ensure the solution supports virtual TOP submissions, so that BCA officers and QPs can carry out remote inspections and measurements. Under the current workflow, BCA reviews a complete virtual inspection submission and returns comments within three working days.

4. Phased implementation timeline

Project roadmap – aligned to the September 2026 FYP cohort

Phase	Activity	Estimated timeline
Phase 1 — Scoping and discovery	Project scoping with academic supervisors, and interviews with project managers and site staff to define robot functions and remote-inspection needs. Student preparation begins mid-August, following examinations.	Aug – Sep 2026
Phase 2 — MVP and simulation (FYP)	The September 2026 FYP cohort runs the partitioned workstreams: simulation-based programming and testing at the PSB campus lab, integrated into a single solution by the end of the project.	Sep 2026 – Apr 2027
Phase 3 — Live pilot trial	Three-month live deployment at the GDC construction site following completion of the FYP; the robot performs daily autonomous site walks.	From May 2027

5. Program structure and student engagement

The initiative is delivered through the Final Year Project (FYP) of PSB Academy's Bachelor of Engineering in Robotics and Artificial Intelligence programme. The FYP runs for roughly eight months; the next cohort begins in September 2026, with student preparation from mid-August following examinations. The integration challenge is partitioned across student teams within that cohort.

- **The “one line” strategy:** rather than assigning the entire integration problem to a single group, we partition it. Each student team owns and solves one piece — for example obstacle avoidance, mechanical mounting, or API integration — and the workstreams are brought together into a single solution by the end of the FYP.
- **Instructional methodology:** we embrace a fail-fast philosophy. The technical success of any single capture is secondary to the learning outcome; students are assessed on how they navigate the development journey and work through complex problems.
- **Industry benchmarking:** OpenSpace experts host open workshops to share how they approach development, giving students a professional-grade benchmark for their own progress.

6. Roles and responsibilities

PSB Academy (institution)	OpenSpace (technology partner)
Student talent: dedicated engineering and robotics cohorts for the partitioned workstreams.	Technology access: 360-degree reality-capture licenses and API access.
Faculty and supervision: academic oversight and project mentorship.	Virtual TOP integration: access to workflows and standards for BCA and QP remote inspections.
Testing grounds: lab facilities and the PSB campus site for Phase 2.	Problem definition: scoping of the commercial and technical hurdles to be solved.
Academic validation: integration into final thesis and viva examinations.	Industry mentorship: expert-led workshops and guidance on global robotics trends.

7. Technical scope of work

To reach a viable autonomous solution, students will address the following:

- **Mechanical mounting:** secure, stable mounts for 360-degree cameras on the robotic platform.
- **Zero-shortcut logic:** programming for total site coverage, preventing the shortcut behaviour common in manual walks, so the record stays complete and trustworthy.
- **Safety features:** auditory sensors that let the robot detect and respond appropriately to site alarms.
- **Battery autonomy:** logic for the robot to recognise low battery and return to a charging dock on its own.
- **Automated data transit:** using the OpenSpace APIs so captured data is uploaded and ready for review within ten minutes of a completed walk.

8. Success metrics and future outlook

We measure success across three horizons:

- **Educational impact:** the number of students trained and the technical quality of their final viva defences. The aim is to produce founders and engineers who can compete with robotics startups in the US and across Asia Pacific.
- **Operational validation:** capture that lets a BCA officer complete a virtual TOP inspection remotely within the current virtual-inspection workflow.

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- **Post-pilot scalability:** extending the technology into facility management — for example mall patrolling, security monitoring, and laboratory-environment tracking.

We intend to formalise the collaboration through a Memorandum of Understanding (MOU), establishing PSB Academy and OpenSpace as a leading partnership for robotics innovation in Singapore — and ensuring our students are equipped to take on the world with world-class technology.